

Quantum Information Theory

Lesson 1: Probably Expected

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Probabilities - Formally

Mnemonic (Probability)

Probability is a relative measure of the set of favourable cases.

Definition (Measures)

Let Ω be a set e.g. $\{X = \text{Heads}, X = \text{Tails}\}$ or $\{X = a \mid a \in [0, 1]\}$.
Let $\mathcal{F}$ be a collection of subsets of Ω that is a σ -algebra (σ -field).
Then a function $\mu : \mathcal{F} \rightarrow \overline{\mathbb{R}}$ (i.e. incl. $\pm\infty$) is a measure iff
 $\mu(E) \geq 0$ for all $E \in \mathcal{F}$ with $\mu(\emptyset) = 0$ and we can add up the
measure of countably many disjoint sets:

$$\mu\left(\bigcup_{i=0}^{\infty} E_i\right) = \sum_{i=0}^{\infty} \mu(E_i) \quad \text{if} \quad E_j \cap E_k = \emptyset \forall j, k.$$

$\mathcal{F}$ being a σ -algebra means that $\Omega \in \mathcal{F}$, that $\mathcal{F}$ is closed under complement i.e. if $E \in \mathcal{F}$ then $\overline{E} = \Omega \setminus E \in \mathcal{F}$, and that $\mathcal{F}$ is closed under countable union i.e. if $\{E_i\} \subseteq \mathcal{F}$ then $\bigcup_{i=0}^{\infty} E_i \in \mathcal{F}$.

μ is a probability measure if, additionally, $\mu(\Omega) = 1$.

Probabilities - Categorical Example

Let's take a semaphore with 3 colors i.e. $\Omega = \{R, Y, G\}$.

To be sure we are closed under all operations, we take $\mathcal{F} = \mathcal{P}(\Omega)$:

$$\mathcal{F} = \{\emptyset, \{R\}, \{Y\}, \{G\}, \{R, Y\}, \{R, G\}, \{Y, G\}, \{R, G, Y\}\}.$$

The simplest measure we can think of is cardinality: $\mu(E) = |E|$.

Note that $|\emptyset| = 0$, $|E_1 \cup E_2| = |E_1| + |E_2| - |E_1 \cap E_2|$ and there are only finitely many disjoint subsets of Ω so this works well.

Cardinality is, however, not a probability measure since $|\Omega| = 3$.

To fix this, we can measure relative cardinality: $P(E) := |E| / |\Omega|$.

However, we usually don't see all outcomes at the same rate.

To fix this, we specify e.g. $\mu(R) = 0.6, \mu(Y) = 0.1, \mu(G) = 0.3$ and work our way from there e.g. $\mu(\{R, G\}) = 0.9$.

Probabilities - Continuous Example

Let's take now a circuit outputting voltage corresponding to a bit. This should be $0V$ when the bit is off and $5V$ when the bit is on. Actually, no. Usually off means $\leq 0.5V$ and on means $\geq 2.7V$. In the continuous case we only care about intervals.

How do we even specify a measure for this kind of space?

Trivially, $\Omega = [0V, 5V]$, but what is $\mathcal{F}$?

Trick: if $\mathcal{T}$ are the open sets of a topological space over Ω , then $\mathcal{T}$ generates a σ -algebra over Ω called the Borel σ -algebra. μ is then completely identified by its restriction restriction to $\mathcal{T}$.

The standard topology on $\mathbb{R}^n$ has the open sets $\mathcal{T}$ be countable unions of open balls $B_r(x_0) = \{x \in \Omega \mid d(x, x_0) < r\}$.

But if we had to specify μ for each such set, no one would do it. It suffices to specify μ for the sets $E_\omega = \{x \in \Omega \mid x \leq \omega\}$.

If we have a continuous increasing bijection $F : [0V, 5V] \rightarrow [0, 1]$, we can define $\mu(\cup_{i=0}^{\infty} (a_i, b_i)) = \sum_{i=0}^{\infty} F(b_i) - F(a_i)$.

F is then the cumulative density function (CDF) associated to μ .

Probabilities - Continuous Example

The simplest cumulative density function we can think of is relative length: $F(x) = \frac{x-0V}{5V-0V}$. There are three caveats however.

Firstly, for us to be able to write this function, the denominator has to be finite. If we need F to be defined on the entire real lines, we need a different shape for F .

Secondly, if we plot F , its graph is a line with slope $1/5$, and yet it is supposed to illustrate an uniform distribution. It is often more useful to plot $f(x) = dF/dx$, the probability density function (PDF), which is a horizontal line at $y = 1/5$. In this new plot, the units on the vertical line have no meaning and can even be above 1.

Lastly, not every voltage has the same probability to occur, e.g. if the circuit is working as specified we should never measure a number between $0.5V$ and $2.7V$. Statistics books are full of distribution types, and physics books show us which of them emerge from certain effects. In this course we discuss neither.

Random Variables & Averages - Formally

We use probabilities to describe the state of a system, as before. Yet, that state is not a probability function, even if described by one. Usually, the state space has its own algebra and measurement unit. The mapping between events and this space is a random variable.

Definition (Random Variables)

Let $(\Omega, \mathcal{F}, \mu)$ be a probability triple as before.

Let (X, Σ) be another σ -algebra (for the values).

A random variable is a function $X : \Omega \rightarrow X$ such that for every subset $A \in \Sigma$ we can calculate the probability of an event occurring leading to one of the values in A , namely

$$\mu(\{X \in A\}) := \mu(\{\omega \mid X(\omega) \in A\}).$$

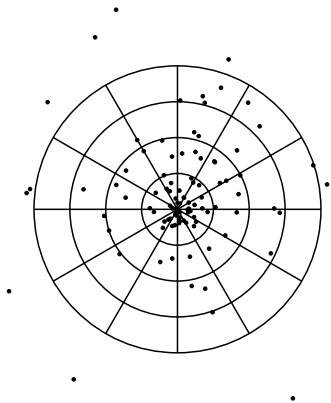
Definition (Averages)

The average (expected value) of X above is the Lebesgue integral

$$\mathbb{E}[X] := \int_{\Omega} X d\mu.$$

Random Variables & Averages - Continuous Example

Shooting at a dartboard, the angles look uniformly distributed, but the radius is not quite halving.

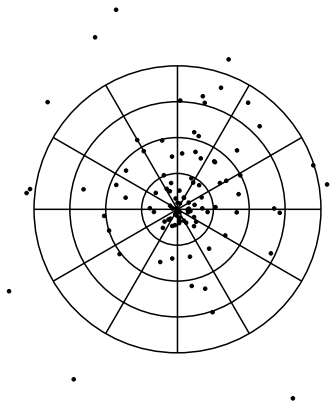


$r_m - r_M$	cnt
0-1	46
1-2	24
2-3	12
3-4	8
4-5	4
5- ∞	6

$\theta_m - \theta_M$	cnt
$\frac{0\pi}{12} - \frac{1\pi}{12}$	10
$\frac{1\pi}{12} - \frac{2\pi}{12}$	10
$\frac{2\pi}{12} - \frac{3\pi}{12}$	12
$\frac{3\pi}{12} - \frac{4\pi}{12}$	10
$\frac{4\pi}{12} - \frac{5\pi}{12}$	7
$\frac{5\pi}{12} - \frac{6\pi}{12}$	8
$\frac{6\pi}{12} - \frac{7\pi}{12}$	6
$\frac{7\pi}{12} - \frac{8\pi}{12}$	6
$\frac{8\pi}{12} - \frac{9\pi}{12}$	8
$\frac{9\pi}{12} - \frac{10\pi}{12}$	8
$\frac{10\pi}{12} - \frac{11\pi}{12}$	6
$\frac{11\pi}{12} - \frac{12\pi}{12}$	9

Random Variables & Averages - Continuous Example

Shooting at a dartboard, the angles look uniformly distributed, and the darts outside the radius r look halving with r .



$r_m - \infty$	cnt
$0 - \infty$	100
$1 - \infty$	54
$2 - \infty$	30
$3 - \infty$	18
$4 - \infty$	10
$5 - \infty$	6

$\theta_m - \theta_M$	cnt
$\frac{0\pi}{12} - \frac{1\pi}{12}$	10
$\frac{1\pi}{12} - \frac{2\pi}{12}$	10
$\frac{2\pi}{12} - \frac{3\pi}{12}$	12
$\frac{3\pi}{12} - \frac{4\pi}{12}$	10
$\frac{4\pi}{12} - \frac{5\pi}{12}$	7
$\frac{5\pi}{12} - \frac{6\pi}{12}$	8
$\frac{6\pi}{12} - \frac{7\pi}{12}$	6
$\frac{7\pi}{12} - \frac{8\pi}{12}$	6
$\frac{8\pi}{12} - \frac{9\pi}{12}$	8
$\frac{9\pi}{12} - \frac{10\pi}{12}$	8
$\frac{10\pi}{12} - \frac{11\pi}{12}$	6
$\frac{11\pi}{12} - \frac{12\pi}{12}$	9

Random Variables & Averages - Continuous Example

Let's say we are throwing darts towards a target board.
The position our dart lands on is a 2D point on the board.
The probability it lands in some region is a number.

Here, it is convenient to describe μ using the sets of balls centered around the center $\mathbf{0} := (0, 0)$. We assume no angular bias and take

$$\mu(\{X \in B_r(\mathbf{0})\}) = 1 - e^{-r/R} = \mu(\{\|X\| < r\}),$$

where R is an accuracy parameter. Note that for $r = 0$ we find probability 0 of hitting the exact bull's eye, and for $r = 1000R$ we still have a (very small) probability of hitting outside that circle.

It's not that hard to compute averages for 1D variables,

$$\mathbb{E}[\|X\|] := \int_{\Omega} \|X\| d\mu = \int_0^{\infty} r \cdot \underbrace{d\left(1 - e^{-r/R}\right)}_{CDF} = \int_0^{\infty} \underbrace{(r/R)e^{-r/R}}_{r \cdot dCDF/dr} dr = R$$

and for 2D we *just* need the formula for $\Pr[X_x < a, X_y < b]$.

Random Variables & Averages - Continuous Example

Let's say we are throwing darts towards a target board.

The position our dart lands on is a 2D point on the board.

We assume no angular bias and let's say

$$\mu(\{X \in B_r(\mathbf{0})\}) = 1 - e^{-r/R} = \mu(\{\|X\| < r\}),$$

and we want to compute the CDF $F(x, y) := \Pr[X_x < a, X_y < b]$.

Note that we have the angular PDF $\frac{d^2}{drd\theta} F(r, \theta) = \frac{1}{2\pi R} e^{-r/R}$.

Consider the rectangle $[x, x + dx] \times [y + dy]$. In polar coordinates, this is not a rectangle $[r, r + dr] \times [\theta + d\theta]$. It's more *like* a parallelogram starting at (r, θ) and with area $1/|J(x, y)| drd\theta$.

We have $r(x, y) = \sqrt{x^2 + y^2}$, $\theta(x, y) = \arctan(y/x) + \frac{\pi}{2} \left(1 - \frac{x}{|x|}\right)$.

The Jacobian of the transformation is (not $r!$; von Mises vs Gauss)

$$|J(x, y)| = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial \theta}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{-y}{x^2 + y^2} \\ \frac{y}{\sqrt{x^2 + y^2}} & \frac{x}{x^2 + y^2} \end{vmatrix} = \frac{1}{\sqrt{x^2 + y^2}}$$

$$\text{so } \frac{d^2}{dx dy} F(x, y) = \frac{d^2}{dr d\theta} F(r(x, y), \theta(x, y)) |J(x, y)| = \frac{1}{2\pi R} \frac{e^{-\frac{1}{R}\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}}.$$

Random Variables & Averages - Continuous Example

Let's say we are throwing darts towards a target board.

The position our dart lands on is a 2D point on the board.

We assume no angular bias and let's say

$$\frac{d^2}{dx dy} F(x, y) = \frac{1}{2\pi R} \frac{e^{-\frac{1}{R}\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}},$$

and we want to compute the marginal PDFs (which are equal)

$$\begin{aligned} \frac{d}{dx} F(x) &= \int_{-\infty}^{\infty} \frac{1}{2\pi R} \frac{e^{-\frac{1}{R}\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}} dy = \frac{1}{2\pi R} \int_{-\infty}^{\infty} \frac{e^{-\sqrt{(x/R)^2+(y/R)^2}}}{\sqrt{(x/R)^2+(y/R)^2}} d(y/R) \\ &= \frac{1}{\pi R} \int_0^{\infty} \frac{e^{-\sqrt{a+z^2}}}{\sqrt{a+z^2}} dz \quad \begin{matrix} u:=\sqrt{a+z^2} \\ \frac{dz}{du}=\frac{u}{\sqrt{u^2-a}} \end{matrix} \frac{1}{\pi R} \int_{\sqrt{a}}^{\infty} \frac{e^{-u}}{\sqrt{u^2-a}} du = K_0\left(\sqrt{\frac{|x|}{R}}\right), \end{aligned}$$

where K_0 , the harmonic Bessel function, is computed numerically.

However, what that means for us is that $\frac{d}{dx} F(x)$ is even so its average is zero. All this tricky math to show that the expected target is the bull's eye, and that's not even because we're hitting it but because we have no angular bias. **We won't do this again.**

Random Variables & Averages - Discrete Example

Let's say we are at a game show, presented with two closed boxes. One of the boxes has $10\times$ as much money as the other.

Let's say the probability that the lowest value box has 10^k \$ is $1/2^k$. After looking at a box, we can choose to switch. Should we? One way to look at this is to compute the switch expected profit.

Let's say we see 100\$. There was 25% chance the boxes were 10\$ and 100\$, and 12.5% chance the boxes were 100\$ and 1000\$.

If we switch we have 66.(6)% chance to lose 90\$ and 33.(3)% chance to win 900\$. Expected profit is thus $-60\$ + 300\$ = 240\$$. If we look at 10^k \$ with $k > 0$, the expected profit is 2.4×10^k \$.

Note that the chance to see 10^k is $50\% \times \frac{1}{2^k} + 50\% \times \frac{1}{2^{k+1}} = \frac{3}{2^{k+2}}$. So, in total the expected profit seems to be

$$2.25 + \sum_{k>0} 2.4 \times 10^k \times \frac{3}{2^{k+2}} = 2.25 + 1.8 \sum_{k>0} 5^k = \infty.$$

Random Variables & Averages - Discrete Example

Let's say we are at a game show, presented with two closed boxes. One of the boxes has $10\times$ as much money as the other.

Let's say the probability that the lowest value box has 10^k \$ is $1/2^k$.

After looking at a box, we can choose to switch. Should we?

At first glance, it looks like, whatever we see, we always should.

So, why bother looking? Just switch.

Since we did not look, we are offered switching again. Should we?

The calculation does not change.

Infinite profit from switching twice (i.e. not at all) as well.

Clearly something is wrong. But what?

Random Variables & Averages - Discrete Example

Let's say we are at a game show, presented with two closed boxes.

One of the boxes has $10\times$ as much money as the other.

Let's say the probability that the lowest value box has $10^k\$$ is $1/2^k$.

After looking at a box, we can choose to switch. Should we?

Let's look at all cases and consider the expected profit.

Without doing any conditioning, for every k there are two cases:
we selected the low box (probability $1/2^{k+1}$, profit 9×10^k) or the
high box (probability $1/2^{k+1}$, profit -9×10^k). Expected profit is

$$\mathbb{E}[P] = \underbrace{9 \times \frac{1}{4}}_{+2.25} - \underbrace{9 \times \frac{1}{4}}_{+9} + \underbrace{90 \times \frac{1}{8} - 90 \times \frac{1}{8}}_{+45} + \underbrace{900 \times \frac{1}{16} - 900 \times \frac{1}{16}}_{+225+\dots} + \dots,$$

but also

$$\mathbb{E}[P] = \underbrace{9 \times \frac{1}{4} - 9 \times \frac{1}{4}}_0 + \underbrace{90 \times \frac{1}{8} - 90 \times \frac{1}{8}}_0 + \underbrace{900 \times \frac{1}{16} - 900 \times \frac{1}{16}}_0 + \dots$$

Random Variables & Averages - Take care with infinities

Let's say we are at a game show, presented with two closed boxes.

One of the boxes has $10\times$ as much money as the other.

Let's say the probability that the lowest value box has 10^k \$ is $1/2^k$.

After looking at a box, we can choose to switch. Should we?

Expected profit is

$$\mathbb{E}[P] = \underbrace{9 \times \frac{1}{4}}_{+2.25} - \underbrace{9 \times \frac{1}{4}}_{+9} + \underbrace{90 \times \frac{1}{8}}_{+45} - \underbrace{90 \times \frac{1}{8}}_{+225+\dots} + \underbrace{900 \times \frac{1}{16}}_{+225+\dots} - \underbrace{900 \times \frac{1}{16}}_{+225+\dots} + \dots,$$

but also

$$\mathbb{E}[P] = \underbrace{9 \times \frac{1}{4} - 9 \times \frac{1}{4}}_0 + \underbrace{90 \times \frac{1}{8} - 90 \times \frac{1}{8}}_0 + \underbrace{900 \times \frac{1}{16} - 900 \times \frac{1}{16}}_0 + \dots$$

The problem is that this series does not converge.

That is why this type of problem creates popular paradoxes.

Profit is a random variable that here does not average.

This is a variant of the, still studied, *two-envelope problem*.

Random Variables & Averages - Categorical Example

Let's say people are betting on a horse race with 6 horses.
The winning horse is an animal's name (or competition number).
The probability that said horse wins is a number in $[0, 1]$.

Here, we specify the probability for each horse number, e.g.

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}$$

which we read as $\Pr[W = A] = 0.1$, $\Pr[W \in \{E, F\}] = 0.5$ etc.

There is no expected winner, because we can't average horses.

Sometimes, categorical data is given number labels.
In that case, you can compute the expected winner number.
However, that number is completely meaningless.

Random Variables & Averages - Finite Example

Let's say people are betting on a horse race with 6 horses.
Let's say they know the probability of each horse winning:

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

When betting, the house tells them how much a 100\$ bet can win.

Horse	A	B	C	D	E	F
Moneyline Odds	800\$	350\$	1,700\$	500\$	200\$	350\$

Let's say that 1,000\$ is bet on each horse (balanced book).

The payout the house gives on this race is a dollar amount.
The probability it gives that payout is a number in $[0, 1]$.

$$P \sim \begin{pmatrix} 9,000\$ & 4,500\$ & 18,000\$ & 6,000\$ & 3,000\$ & 4,500\$ \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

From this we can calculate the expected payout and profit.

Random Variables & Averages - Finite Example

Let's say people are betting on a horse race with 6 horses.
Let's say that 1,000\$ is bet on each horse (balanced book).
The payout the house gives on this race is a dollar amount.

$$P \sim \begin{pmatrix} 9,000\$ & 4,500\$ & 18,000\$ & 6,000\$ & 3,000\$ & 4,500\$ \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

From this we can calculate the expected payout and profit.

$$\begin{aligned} \mathbb{E}[P] &:= \sum_i x_i \Pr[P = x_i] = 0.1 \times 9,000\$ + 0.2 \times 4,500\$ \\ &\quad + 0.05 \times 18,000\$ + 0.15 \times 6,000\$ \\ &\quad + 0.3 \times 3,000\$ + 0.2 \times 4,500\$ = 6 \times 900\$. \end{aligned}$$

Neat how the house had taken in $6 \times 1,000\$$ right?

Random Variables & Averages - Finite Example

Let's say we are bookkeeping fixed odds for a race with n horses. Let's say we expect a balanced book, e.g. 1\$ bet on each horse. We have an estimate of each horse winning, $\tilde{p}_i \approx \Pr[W = h_i]$. Our job is to rig the odds to make sure people lose 10%.

So, we have to engineer a random variable

$$\tilde{O} \sim \begin{pmatrix} o_1 \times & o_2 \times & \dots & o_n \times \\ \tilde{p}_1 & \tilde{p}_2 & \dots & \tilde{p}_n \end{pmatrix}$$

such that $0.9n \times = \mathbb{E}[\tilde{O}] := \sum_i o_i \tilde{p}_i \times$. In theory, there are many. A trivial solution is to pick $o_i = \frac{0.9 \times}{\tilde{p}_i}$. All terms become $0.9 \times$. The quantity $1/\tilde{p}_i$ is known as the (prior) odds of the event.

Of course, we did not know the exact $p_i = \Pr[W = h_i]$.

The average payout is $\mathbb{E}[\tilde{O}] = \sum_i \frac{0.9 p_i \times}{\tilde{p}_i} \in \left[0.9n \times, 0.9n \max_i \frac{p_i}{\tilde{p}_i} \times\right]$.

Also, we did not know the winner. The realized payout is $\frac{0.9 \times}{p_W}$.

Random Variables & Averages - Finite Example

Let's say we are bookkeeping fixed odds for a race with n horses. We have an estimate of each horse winning, $\tilde{p}_i \approx \Pr[W = h_i] = p_i$. Our job is to rig the odds to make sure people lose a fraction f . For a 1\$ bet on h_i we will pay out $o_i = \frac{1-f}{\tilde{p}_i}$.

Let's say we expect b_i \$ bet on each horse h_i . The actual payout is

$$P \sim \begin{pmatrix} (1-f)\frac{b_1}{\tilde{p}_1} \$ & (1-f)\frac{b_2}{\tilde{p}_2} \$ & \dots & (1-f)\frac{b_n}{\tilde{p}_n} \$ \\ p_1 & p_2 & \dots & p_n \end{pmatrix},$$

so on average

$$\mathbb{E}[P] = \sum_i (1-f) \frac{b_i}{\tilde{p}_i} p_i \$ \leq \underbrace{(1-f)}_{\text{odds key}} \underbrace{\sum_i b_i}_{\text{predicted take-in}} \underbrace{\max_i \frac{p_i}{\tilde{p}_i}}_{\text{odds quality}} \$.$$

Notice that if we are not expecting a balanced book, we may wish to amend our odds, weighing b_i into o_i . That's why odds change.

Random Variables & Averages - Finite Example

Let's say we are bookkeeping odds for a race with n horses.

Let's say there is already a L_i liability on each horse h_i , accrued by bets B_i being placed at possibly different odds in the past.

We have an estimate of each horse winning, $\tilde{p}_i \approx \Pr[W = h_i] = p_i$.

We have an estimate of the remaining bets on each horse $\tilde{b}_i$.

Our job is to rig the odds to make sure people lose a fraction f .

So, we have to engineer a random variable

$$\tilde{p} \sim \begin{pmatrix} L_1 + \tilde{b}_1 o_1 \$ & L_2 + \tilde{b}_2 o_2 \$ & \dots & L_n + \tilde{b}_n o_n \$ \\ \tilde{p}_1 & \tilde{p}_2 & \dots & \tilde{p}_n \end{pmatrix}$$

$$\text{such that } (1-f) \underbrace{\sum_i B_i + \tilde{b}_i}_{\text{predicted take-in}} = \mathbb{E}[\tilde{P}] = \sum_i \underbrace{(L_i + \tilde{b}_i o_i)}_{\text{predicted liability}} \tilde{p}_i.$$

There are now two easy solutions:

- ▶ $L_i + \tilde{b}_i o_i = \frac{1-f}{\tilde{p}_i} (B_i + \tilde{b}_i)$: you lose when the fans win – **risky**.
- ▶ $L_i + \tilde{b}_i o_i = \frac{1-f}{\tilde{p}_i} \frac{\sum_i B_i + \tilde{b}_i}{n}$: the house wins every time – **good**.

Random Variables & Averages - Finite Example

Let's say we are bookkeeping odds for a race with n horses.

Let's say there is already a L_i liability on each horse h_i , accrued by bets B_i being placed at possibly different odds in the past.

We have an estimate of each horse winning, $\tilde{p}_i \approx \Pr[W = h_i] = p_i$.

We have an estimate of the remaining bets on each horse $\tilde{b}_i$.

For a bet on h_i we will be paying o_i such that

$$L_i + \tilde{b}_i o_i = \frac{1 - f}{\tilde{p}_i} \frac{\sum_i B_i + \tilde{b}_i}{n}.$$

Let's say we predict 1,200 bets of 1\$ each on a fair two-horse race.

To keep 10% we offer odds $o_1 = o_2 = 0.9/0.5 = 1.8\times$.

Yet, after 800 bets come in, we see that 600 were all on h_1 .

We have $B_1 = 600\$$, $L_1 = 1,080\$$, $B_2 = 200\$$, $L_2 = 360\$$.

It is fair to think that $\tilde{b}_1 = 300\$$, $\tilde{b}_2 = 100\$$. So we adjust

$$\blacktriangleright 1,080\$ + 300\$ o_1 = 1.8 \frac{1200\$}{2} \Rightarrow o_1 = \frac{1080\$ - 1080\$}{300\$} = 0,$$

$$\blacktriangleright 360\$ + 100\$ o_2 = 1.8 \frac{1200\$}{2} \Rightarrow o_2 = \frac{1080\$ - 360\$}{100\$} = 7.2\times \gg 1\times,$$

i.e. we shouldn't offer any more bets on h_1 and we should offer astronomic odds on h_2 because if h_1 wins we need to pay that.

Random Variables & Averages - Finite Example

Let's say we are bookkeeping odds for a race with n horses.

Let's say there is already a L_i liability on each horse h_i , accrued by bets B_i being placed at possibly different odds in the past.

We have an estimate of each horse winning, $\tilde{p}_i \approx \Pr[W = h_i] = p_i$.

We have an estimate of the remaining bets on each horse $\tilde{b}_i$.

For a bet on h_i we will be paying o_i such that

$$L_i + \tilde{b}_i o_i = \frac{1 - f}{\tilde{p}_i} \frac{\sum_i B_i + \tilde{b}_i}{n}.$$

Let's say we predict 1,000 bets of 1\$ each on a fair two-horse race.

To keep 10% we offer odds $o_1 = o_2 = 0.9/0.5 = 1.8\times$.

Yet, after 800 bets come in, we see that 600 were all on h_1 .

We have $B_1 = 600\$$, $L_1 = 1,080\$ > 1000\$$, $B_2 = 200\$$, $L_2 = 360\$$.

It is fair to think that $\tilde{b}_1 = 150\$$, $\tilde{b}_2 = 50\$$. So we adjust

$$\blacktriangleright 1,080\$ + 150\$ o_1 = 1.8 \frac{1000\$}{2} \Rightarrow o_1 = \frac{900\$ - 1080\$}{150\$} = -1.2\times < 0,$$

$$\blacktriangleright 360\$ + 50\$ o_2 = 1.8 \frac{1000\$}{2} \Rightarrow o_2 = \frac{900\$ - 360\$}{50\$} = 10.8\times \gg 1\times,$$

i.e. we need to be paid to take bets on h_1 and we offer desperate odds on h_2 because if h_1 wins we need to pay that.

Follow-up Reading

- ▶ C.Bishop – Pattern Recognition and Machine Learning Springer (2006), Ch. 1,2: Intro & Probability Distributions
<http://www.cs.man.ac.uk/~fumie/tmp/bishop.pdf>
- ▶ C.Niculescu – Probabilități și Statistică Ed. Univ. din București (2015), Ch. 1,2,6: σ -fields, Random Variables, n -dimensional variables, transformation's Jacobian
- ▶ B.E.Fristedt, L.F.Gray – A Modern Approach to Probability Theory, Springer (1996), Ch. 1-10: **Everything. Detailed.**
- ▶ M.D.McDonnell, A.Derek – Randomized switching in the two-envelope problem, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 465.2111 (2009), pp. 3309-3322: two-envelope problem.